WORKED SOLUTION

STEP III 1998, Question 1

Pure • Curve Sketching

STEP III 1998, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q1

The key idea

Write everything in terms of $\cos x$: with $\sin^2 x = 1 - \cos^2 x$, the function f becomes a quadratic in $\cos x$. Completing the square tells us the range of f at a glance, and this range is what controls the final part. For g, the quotient rule gives a derivative that is a constant multiple of $\mathrm{f}'(x)$, divided by a square.

Solution

Sketching $y = \mathrm{f}(x)$

Using $\sin^2 x = 1 - \cos^2 x$,

$$\mathrm{f}(x) = 2 + 2\cos x - \cos^2 x = 3 - (1 - \cos x)^2.$$

Since $0 \leq 1 - \cos x \leq 2$, we get $-1 \leq \mathrm{f}(x) \leq 3$. The value 3 is attained when $\cos x = 1$ and the value -1 when $\cos x = -1$.

Differentiating,

$$\mathrm{f}'(x) = -2(1 - \cos x)\sin x = 2\sin x\,(\cos x - 1).$$

This is zero when $\sin x = 0$ or $\cos x = 1$, which on $[0, 2\pi]$ means $x = 0,\ \pi,\ 2\pi$. The stationary points are

$$(0,\,3), \qquad (\pi,\,-1), \qquad (2\pi,\,3).$$

On $(0, \pi)$ we have $\sin x > 0$ and $\cos x - 1 < 0$, so $\mathrm{f}' < 0$ and f is decreasing. On $(\pi, 2\pi)$, $\sin x < 0$ while $\cos x - 1$ is still negative, so $\mathrm{f}' > 0$ and f is increasing. So $(\pi, -1)$ is a minimum, and the end points $(0, 3)$ and $(2\pi, 3)$ are maxima.

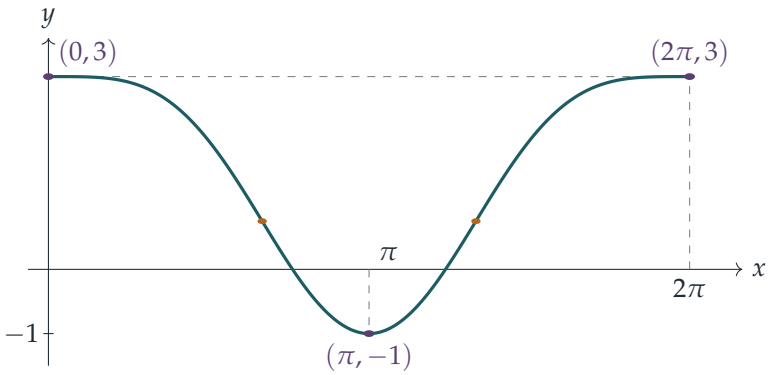
? **Ask yourself** How does the curve look near $x = 0$? Is it like a parabola?

No, it's flatter than a parabola. Near $x = 0$, $1 - \cos x \approx \frac{1}{2}x^2$, so $\mathrm{f}(x) \approx 3 - \frac{1}{4}x^4$. The curve is very flat at the top, which is consistent with $\mathrm{f}''(0) = 0$. Near $x = \pi$, write

$x = \pi + t$. Then $1 - \cos x = 1 + \cos t \approx 2 - \frac{1}{2}t^2$, so $\mathrm{f} \approx -1 + 2t^2$: an ordinary parabolic minimum.

Some other useful features for the sketch:

- The curve is symmetric about $x = \pi$, since $\cos(2\pi - x) = \cos x$.
- It crosses the x-axis where $(1 - \cos x)^2 = 3$, that is, $\cos x = 1 - \sqrt{3}$. This gives $x = \arccos(1 - \sqrt{3}) \approx 2.39$ and $x = 2\pi - \arccos(1 - \sqrt{3}) \approx 3.89$.
- $\mathrm{f}''(x) = 2\cos 2x - 2\cos x = 2(2\cos x + 1)(\cos x - 1)$. This changes sign where $\cos x = -\frac{1}{2}$, so the points of inflection are at $\left(\frac{2\pi}{3}, \frac{3}{4}\right)$ and $\left(\frac{4\pi}{3}, \frac{3}{4}\right)$.



The amber dots are the points of inflection.

Stationary points of g

By the quotient rule, wherever $c\,\mathrm{f}(x) + d \neq 0$,

$$\mathrm{g}'(x) = \frac{a\mathrm{f}'(x)\big(c\mathrm{f}(x) + d\big) - c\mathrm{f}'(x)\big(a\mathrm{f}(x) + b\big)}{\big(c\mathrm{f}(x) + d\big)^2} = \frac{(ad - bc)\,\mathrm{f}'(x)}{\big(c\mathrm{f}(x) + d\big)^2}.$$

Since $ad \neq bc$, we have $\mathrm{g}'(x) = 0$ exactly when $\mathrm{f}'(x) = 0$, at every point where g is defined. So g has no stationary points other than those of f.

We still need g to be defined at $x = 0, \pi, 2\pi$, which is where the other two conditions come in. At $x = 0$ and $x = 2\pi$ the denominator is $3c + d$, which is non-zero because $d \neq -3c$. At $x = \pi$ it is $d - c$, which is non-zero because $d \neq c$. So the stationary points of $y = \mathrm{g}(x)$ are at exactly $x = 0,\ \pi,\ 2\pi$, as required, with

$$\mathrm{g}(0) = \mathrm{g}(2\pi) = \frac{3a + b}{3c + d}, \qquad \mathrm{g}(\pi) = \frac{b - a}{d - c}.$$

When $|\mathrm{g}|$ is bounded

Suppose $d/c < -3$ or $d/c > 1$. In particular $c \neq 0$, so we can write

$$c\,\mathrm{f}(x) + d = c\left(\mathrm{f}(x) + \frac{d}{c}\right).$$

As $\mathrm{f}(x)$ takes values in $[-1,3]$, the bracket lies in $\left[\frac{d}{c}-1,\ \frac{d}{c}+3\right]$.

- If $d/c > 1$, the bracket is at least $\frac{d}{c}-1>0$.
- If $d/c < -3$, the bracket is at most $\frac{d}{c}+3<0$.

Either way there is a constant $k>0$ with $\left|c\,\mathrm{f}(x)+d\right| \geq |c|\,k$ for all x. In words, the denominator stays away from zero. The numerator is bounded, because $|a\mathrm{f}(x)+b| \leq 3|a|+|b|$. Hence

$$|\mathrm{g}(x)| \leq \frac{3|a|+|b|}{|c|\,k}$$

for every x in $[0,2\pi]$, so $|\mathrm{g}(x)|$ cannot be arbitrarily large.

Ask yourself What goes wrong if $-3 < d/c < 1$?

Then $-d/c$ lies strictly between -1 and 3. Since f is continuous and takes every value in $[-1,3]$, there is an x with $\mathrm{f}(x) = -d/c$, and the denominator vanishes there. The numerator does not vanish at the same point: if both did, we would have $a(-d/c)+b = 0$, that is, $ad = bc$. So g has a vertical asymptote, and $|\mathrm{g}|$ is unbounded. This shows that the condition in the final part is exactly the right one.

Checks

- Numerically, f over a fine grid on $[0,2\pi]$ has minimum -1 and maximum 3. Also $3-\mathrm{f}(0.1) \approx 2.496\times10^{-5}$, against $\frac{1}{4}(0.1)^4 = 2.5\times10^{-5}$, confirming the quartic flatness at $x=0$.
- A special case: $a=1$, $b=0$, $c=0$, $d=1$ gives $\mathrm{g}=\mathrm{f}$, and the formulas give $\mathrm{g}(0)=3$ and $\mathrm{g}(\pi)=-1$, as they should.
- For random a,b,c,d, a finite-difference derivative of g agrees with $(ad-bc)\mathrm{f}'/(c\mathrm{f}+d)^2$.

Common slips

- Missing the end points $x=0$ and $x=2\pi$ as stationary points. f' really is zero there.
- Forgetting to check that g is defined at the stationary points of f. That is exactly what the conditions $d \neq -3c$ and $d \neq c$ are for.
- Drawing a sharp, parabola-shaped top at $x=0$. The curve is much flatter there.

Beyond STEP

The map $t \mapsto \dfrac{at+b}{ct+d}$ with $ad \neq bc$ is a *Möbius transformation*. It is invertible, and its derivative $(ad-bc)/(ct+d)^2$ never vanishes, which is why composing with it cannot

create or destroy stationary points. Möbius maps are central in complex analysis and geometry, and they form a group under composition. For a beautiful picture-led account, see Needham, *Visual Complex Analysis*.

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